Green function derivation of trench MOSFET breakdown

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This report replaces the transverse Taylor approximation with the Green function of a specified two-dimensional Poisson boundary problem. It derives the kernel, integrates the donor and drain contributions, implements a rapidly convergent equivalent series, and recalculates the breakdown curves. No coefficients are fitted to the source curves. The resulting experimental-geometry centerline avalanche voltage is **34.27 V**, compared with 34.85 V from the previous approximation and 35.4 V reported experimentally.

First page nomenclature

Symbol	Definition	Unit
x, y	Downward and transverse coordinates	cm
ξ, η	Green function source coordinates	cm
$L, a = W_s$	Drift length and silicon half width	cm
W_I, b	Oxide thickness; $b = \varepsilon_s W_I/\varepsilon_I$	cm
N_d, q	Ionized donor density; elementary charge	cm^{-3}; C
$\varepsilon_s, \varepsilon_I$	Absolute permittivities	F/cm
$k = qN_d/\varepsilon_s$	Uniform source in depleted silicon	$\mathrm{V/cm}^2$
V, ϕ	Drain voltage and electrostatic potential	V
$G(P, Q)$	Green function for $-\nabla^2$	1
X_m, D_m, κ_m	Transverse mode, norm, eigenvalue	1; cm; cm^{-1}
$g_m(x, \xi)$	One-dimensional mode Green function	cm
$\mu_n = n\pi/L$	Longitudinal sine wavenumber	cm^{-1}
$R_n(y)$	Transverse transfer factor	1
f, h	Basis potentials in $\phi = Vf + kh$	1; cm^2
$E_+ = \phi_x(x, 0)$	Positive centerline field convention	V/cm
α_n, α_p	Electron and hole ionization coefficients	cm^{-1}
I_n, I_p	Avalanche functionals	1
V_{BD}	Centerline avalanche voltage in this model	V

Lengths stated in μm are converted to cm before computation. The physical field vector is $\mathbf{E} = -\nabla\phi$; therefore positive E_+ corresponds to electron drift in the $+x$ direction. Temperature is fixed at 300 K for the chosen ionization coefficients.

Scope. Electrostatics is exact for the stated rectangular Robin benchmark in the infinite-series limit. The finite-mode computation is numerical. The oxide approximation, device geometry and one-path avalanche model remain physical approximations; global corner breakdown and partial depletion are not solved.

1 Geometry and the boundary value problem

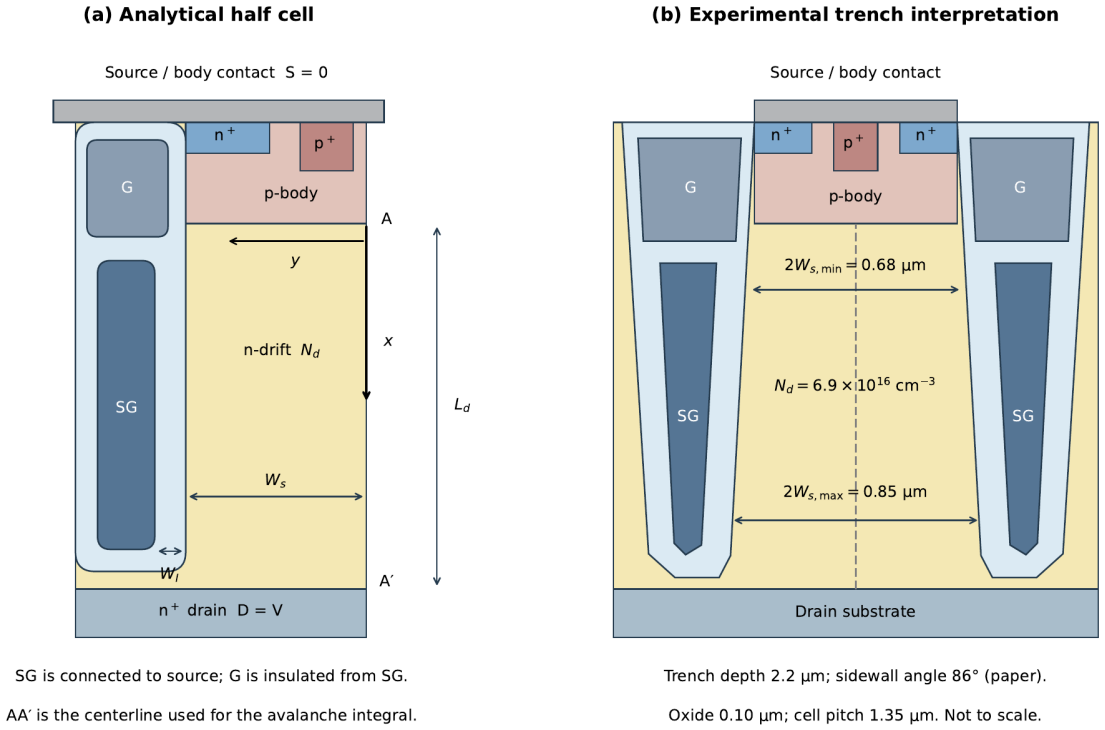


Figure 1: Shield gate trench interpretation based on Zhang et al. [2, Figs. 5(a), 7(a) and Table I]. The exact Green function is applied to the rectangular analytical half cell, not to the tapered experimental shape on the right. Drawings are conceptual and not to scale.

The upper gate and lower source-connected shield gate are separate electrodes. Let $\Omega = (0, L) \times (0, a)$ with centerline $y = 0$. In depleted silicon,

$$-\nabla^2 \phi = k, \qquad k = \frac{qN_d}{\varepsilon_s}.$$

The specified boundary conditions are

$$\phi(0, y) = 0, \quad \phi(L, y) = V, \quad \phi_y(x, 0) = 0, \quad \phi_y(x, a) + \frac{\phi(x, a)}{b} = 0.$$

The source/body boundary is now grounded across the full top edge. This is an additional idealization relative to the centerline-only top condition in the second paper's reduced derivation. The oxide is replaced by a local capacitance; fixed interface charge and work-function offsets are omitted. Donors are uniform and fully ionized. Built-in voltages, body-profile details and mobile charge in the depleted region are omitted.

For the experimental comparison, use $a = 0.425$ μm, $W_I = 0.10$ μm, $L = 1.25$ μm and $N_d = 6.9 \times 10^{16}$ cm^{-3}. The physical mesa tapers between half widths 0.34 and 0.425 μm; this calculation uses a constant 0.425 μm. The trench depth 2.2 μm and drift length 1.25 μm are not interchangeable.

The two plotted geometry families are $(W_I, L) = (0.25, 1.6)$ μm and $(0.40, 3.0)$ μm, with $a = 0.2, 0.3, 0.5, 0.7, 0.9$ μm. The 0.2-μm member of the first family is an added sweep. Use $\varepsilon_s = 11.7\varepsilon_0$, $\varepsilon_I = 3.9\varepsilon_0$, $\varepsilon_0 = 8.8541878128 \times 10^{-14}$ F/cm and $q = 1.602176634 \times 10^{-19}$ C.

2 Construction of the Green function

Let $P = (x, y)$ and $Q = (\xi, \eta)$. Define the Green function by

$$-\nabla_P^2 G(P, Q) = \delta(x - \xi)\delta(y - \eta),$$

with homogeneous versions of all four boundary conditions. Separate transverse eigenfunctions:

$$X_m(y) = \cos(\kappa_m y), \qquad \kappa_m \tan(\kappa_m a) = \frac{1}{b}, \qquad m\pi < \kappa_m a < m\pi + \frac{\pi}{2}.$$

Their norms and completeness relation are

$$D_m = \int_0^a X_m^2 dy = \frac{a}{2} + \frac{\sin(2\kappa_m a)}{4\kappa_m}, \qquad \sum_{m=0}^{\infty} \frac{X_m(y)X_m(\eta)}{D_m} = \delta(y - \eta).$$

For each mode, require

$$\left(-\frac{d^2}{dx^2} + \kappa_m^2\right) g_m(x, \xi) = \delta(x - \xi), \qquad g_m(0, \xi) = g_m(L, \xi) = 0.$$

The solutions on either side of the source are proportional to $\sinh(\kappa_m x)$ and $\sinh[\kappa_m(L-x)]$. Continuity and the derivative jump

$$\partial_x g_m(\xi^+, \xi) - \partial_x g_m(\xi^-, \xi) = -1$$

fix the normalization. With $x_< = \min(x, \xi)$ and $x_> = \max(x, \xi)$,

$$g_m(x, \xi) = \frac{\sinh(\kappa_m x_<)\sinh[\kappa_m(L - x_>)]}{\kappa_m \sinh(\kappa_m L)}.$$

Therefore the required two-dimensional kernel is

$$G(P, Q) = \sum_{m=0}^{\infty} \frac{X_m(y)X_m(\eta)}{D_m} \frac{\sinh(\kappa_m x_<)\sinh[\kappa_m(L - x_>)]}{\kappa_m \sinh(\kappa_m L)}.$$

The kernel is reciprocal: $G(P, Q) = G(Q, P)$. Applying $-\nabla_P^2$ gives the normalized delta source by completeness. Its singularity at $P = Q$ is understood in the distributional sense; source integration is performed analytically below. The full 2D Green function is dimensionless because g_m/D_m is dimensionless.

3 Green representation and analytic source integration

Green's second identity for the operator $-\nabla^2$ gives

$$\phi(P) = k \int_\Omega G(P,Q)\, dQ - \int_{\Gamma_D} \phi(Q) \partial_{n_Q} G(P,Q)\, dS_Q.$$

The Robin boundary term cancels because G and ϕ obey the same homogeneous Robin relation. The symmetry boundary also contributes zero. At the top $\phi = 0$; only the bottom value V remains. Hence

$$\phi(P) = k \int_0^L \int_0^a G(P,Q)\, d\eta\, d\xi - V \int_0^a \partial_\xi G(P,Q)|_{\xi=L}\, d\eta.$$

The negative sign is essential: the outward normal at the drain points in $+\xi$. Define the constant-expansion coefficient

$$c_m = \frac{1}{D_m} \int_0^a X_m(\eta)\, d\eta = \frac{\sin(\kappa_m a)}{\kappa_m D_m}.$$

The mode integrations evaluate to

$$\int_0^L g_m(x,\xi)\, d\xi = \frac{1}{\kappa_m^2} \left[1 - \frac{\cosh[\kappa_m (x - L/2)]}{\cosh(\kappa_m L/2)} \right],$$

$$- \partial_\xi g_m(x,\xi)|_{\xi=L} = \frac{\sinh(\kappa_m x)}{\sinh(\kappa_m L)}.$$

Thus the potential is the explicit eigenfunction series

$$\phi(x,y) = \sum_{m=0}^{\infty} c_m \cos(\kappa_m y) \Bigg\{ V \frac{\sinh(\kappa_m x)}{\sinh(\kappa_m L)} + \frac{k}{\kappa_m^2} \left[1 - \frac{\cosh[\kappa_m (x - L/2)]}{\cosh(\kappa_m L/2)} \right] \Bigg\}.$$

This is the same benchmark potential obtained previously by separation of variables, now derived through a normalized Green kernel and its boundary representation. The construction explains both the source normalization and the drain-boundary sign. It is not a finite elementary closed form.

Uniqueness follows by applying the energy identity to the difference u of two solutions: $\int_\Omega |\nabla u|^2 dA + b^{-1} \int_{y=a} u^2 dx = 0$, with zero end traces. Consequently $u = 0$.

4 Rapidly convergent evaluation at the centerline

For numerical evaluation, use an equivalent longitudinal sine expansion. It avoids the slow derivative convergence of the transverse series near the drain. Choose the lifting potential

$$P(x) = \frac{Vx}{L} + \frac{k}{2}x(L-x), \qquad -P'' = k.$$

Its sine coefficients on $0 < x < L$ are

$$P_n = Vv_n + kw_n, \qquad v_n = \frac{2(-1)^{n+1}}{n\pi}, \qquad w_n = \frac{2L^2[1-(-1)^n]}{(n\pi)^3}.$$

Set $\mu_n = n\pi/L$. A harmonic correction satisfying symmetry and cancelling the sidewall residual is obtained with

$$R_n(y) = \frac{\cosh(\mu_n y)}{\cosh(\mu_n a) + b\mu_n \sinh(\mu_n a)}.$$

It satisfies $R_n'(0) = 0$ and $R_n'(a) + R_n(a)/b = 1/b$. Therefore

$$\phi(x,y) = P(x) - \sum_{n=1}^{\infty} P_n R_n(y) \sin(\mu_n x).$$

Each correction term is harmonic and vanishes at both end traces. On the open sidewall its Robin residual cancels $P(x)/b$ by the sine expansion. Uniqueness establishes equality to the Green representation. No drain correction coefficient or fitted effective thickness is needed for this specified boundary problem.

At $y = 0$ the field used in the avalanche integral is

$$E_+(x) = \frac{V}{L} + k\left(\frac{L}{2} - x\right) - \sum_{n=1}^{\infty}(Vv_n + kw_n)\mu_n R_n(0)\cos(\mu_n x).$$

The stable transfer factor used in code is

$$R_n(y) = \frac{e^{-\mu_n(a-y)} + e^{-\mu_n(a+y)}}{(1+b\mu_n) + (1-b\mu_n)e^{-2\mu_n a}}.$$

At the centerline its exponential decay makes potential and field sums rapidly convergent. Convergence is slower at the sidewall and singular corner; centerline convergence does not certify a global maximum field. The code uses 240 modes for curves and 480 for the displayed potential map.

5 Avalanche criterion and domain of admissibility

Write $\phi = Vf + kh$ and $E_+ = Vf_x + kh_x$. Both f and h are computed from the preceding series. On the sampled centerline require $V(1-f) - kh \geq 0$ and $E_+ \geq 0$. Include the bottom limit $h/(1-f) \to -h_x(L,0)/f_x(L,0)$. These tests reject field reversal and potential overshoot along the assumed path; they do not prove full depletion of the entire device.

Choose the original van Overstraeten–de Man coefficients at 300 K [3,4], with fields in V/cm and coefficients in cm^{-1}:

$$\alpha_n(F) = 7.03 \times 10^5 e^{-1.231\times 10^6/F},$$

$$\alpha_p(F) = \begin{cases} 1.582 \times 10^6 e^{-2.036\times 10^6/F}, & F < 4\times 10^5, \\ 6.71 \times 10^5 e^{-1.693\times 10^6/F}, & F \geq 4\times 10^5. \end{cases}$$

Here $F = E_+ \geq 0$ on the valid branch. The zero-field limit is zero. With electron drift in $+x$, define

$$C(x) = \int_0^x (\alpha_p - \alpha_n)\, dt, \qquad I_n(V, N_d) = \int_0^L \alpha_n(x) e^{C(x)}\, dx.$$

Solve $I_n(V_{\mathrm{BD}}, N_d) = 1$ by bracketing and Brent root finding. The corresponding hole functional is $I_p = \int_0^L \alpha_p(x) e^{C(x)-C(L)} dx$. The identity $I_n - 1 = e^{C(L)}(I_p - 1)$ gives an independent numerical check. For equal constant coefficients both reduce to αL.

The sweep uses 2001 longitudinal integration points and 251 logarithmically spaced densities from 10^{15} to 1.6×10^{17} cm^{-3}. Starting voltage is the largest of zero, the sampled no-overshoot requirement and the no-reversal requirement. If the avalanche integral is already at least unity there, no voltage from this branch is plotted. An upper voltage bracket is expanded until the integral exceeds unity; root tolerance is 10^{-9} V. Quadrature convergence, not the root tolerance, sets numerical reliability.

These material coefficients are an explicit choice, not recovered simulator settings from [2]. No parameters are adjusted to its curves. The Green function determines electrostatics; numerical integration and a numerical root are still needed for avalanche. The result is an analytical electrostatic model with a numerical centerline breakdown prediction, not a closed-form law for physical global breakdown.

6 Recalculated breakdown curves

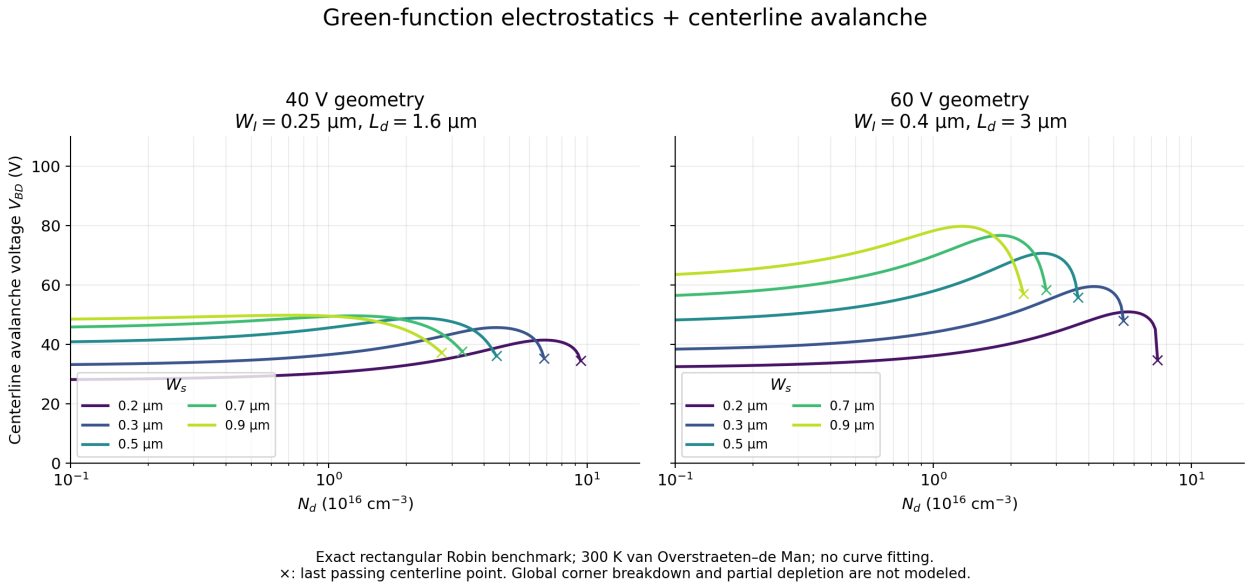


Figure 2: Green function prediction for the second paper's geometry families. Each cross is the last sampled point passing the centerline tests. No physical partial-depletion or lateral-avalanche branch is extrapolated.

The curves retain the rise, maximum and initial decline. Applied voltage and donor fields redistribute the longitudinal field, and their relative contributions change with doping. The mathematical method is different from the earlier Taylor-based model, while the geometry and avalanche-law assumptions are held fixed for comparison.

Half width (μm)	40-V family peak N_d (10^{16} cm^{-3})	Peak voltage (V)	60-V family peak N_d (10^{16} cm^{-3})	Peak voltage (V)
0.2	6.821	41.44	5.682	50.94
0.3	4.453	45.65	4.190	59.48
0.5	2.233	48.83	2.627	70.67
0.7	1.265	49.61	1.823	76.70
0.9	0.716	49.81	1.291	79.76

These are sampled maxima, not continuously optimized design points. The family names come from the paper and are not constraints forcing the calculated voltage to be 40 or 60 V. The second paper's entire Fig. 3 has not been quantitatively validated by these curves; they also do not reconstruct the earlier paper's Fig. 10, whose physical geometry remains unidentified.

7 Potential map and experimental comparison

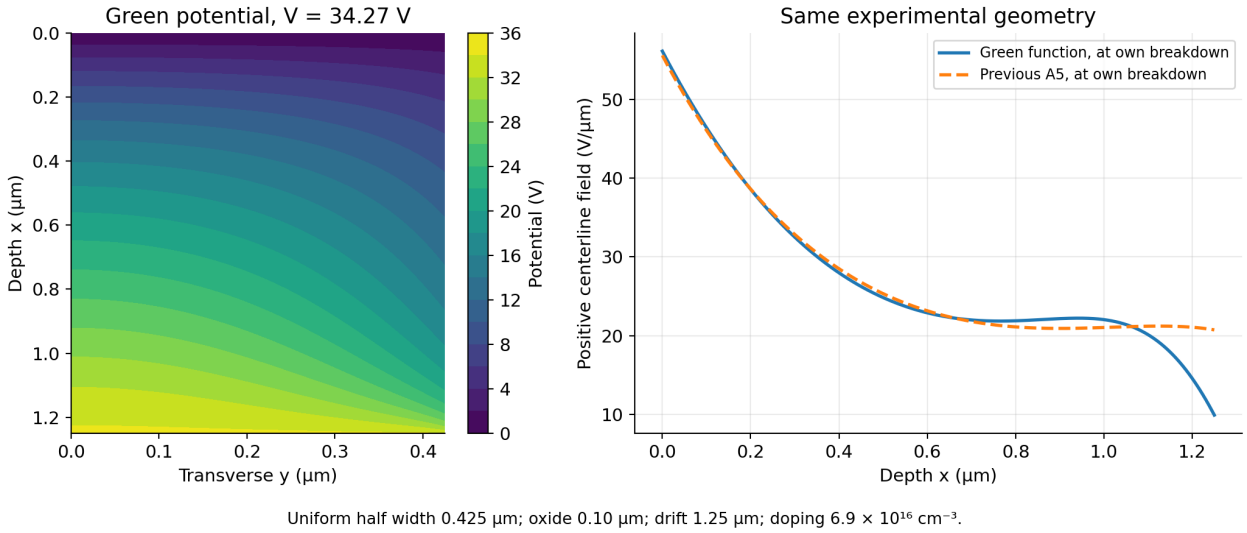


Figure 3: Computed two-dimensional Green potential and centerline field for the experimental parameter set. The dashed comparison is the previous A5 approximation. Each field curve is evaluated at its own centerline avalanche voltage, not at a common applied voltage.

Model or observation	Voltage (V)	Difference from 35.4 V
Green function rectangular benchmark	34.2742	−3.18%
Previous Taylor and drain-correction model A5	34.8542	−1.54%
Experiment, Section IV at 100 mA	35.4	—
Experiment, Table I	35.5	+0.28%

The Green result is about 0.580 V below the previous calculation. It is not closer to the measured value. Greater mathematical accuracy within an ideal boundary model does not guarantee greater physical accuracy for a tapered device with detailed junctions and finite-current measurement criteria.

At the Green solution, the centerline minimum field is 9.9365×10^4 V/cm and the numerical voltage integral is 34.274208536 V. On the displayed 181-by-91 potential grid, the maximum of $\phi - V$ is zero, attained at the drain boundary. This sampled check is consistent with no potential overshoot for this particular operating point; it is not a proof covering every geometry and density in the plot.

The same constant 0.425-μm half width is used in both models. Neither reproduces the measured taper from 0.34 to 0.425 μm. The comparison does not tune the geometry or avalanche coefficients to the experimental value.

8 Independent verification and convergence

8.1 Finite difference comparison

A separately assembled five-point Poisson matrix uses the same end potentials, centered ghost-node symmetry and Robin boundary conditions. At the experimental parameter set and $V = 34.274208536$ V, compare four points $(x/L, y/a) = (0.25, 0), (0.5, 0), (0.75, 0.5), (0.5, 1)$ with the Green solution.

Grid (n_x, n_y)	Maximum sampled potential difference (V)
(61, 21)	1.24318×10^{-3}
(121, 41)	3.11587×10^{-4}
(241, 81)	7.79467×10^{-5}

The approximately fourfold reduction under mesh halving supports second-order agreement. This is a sampled comparison, not a global error norm. A second evaluation using 1600 transverse Green modes agrees with the 640-mode longitudinal representation at these points to 3.53×10^{-5} V; finite sidewall truncation is included in that difference.

8.2 Avalanche integration refinement

Integration points	40-V test (V)	60-V test (V)	Experimental geometry (V)
1001	47.604090	69.721165	34.273913
2001	47.604169	69.721590	34.274138
4001	47.604189	69.721697	34.274194
8001	47.604193	69.721723	34.274209

The first two tests use $a = 0.5$ μm and $N_d = 3 \times 10^{16}$ cm^{-3} in their respective families. The 2001-to-8001 point changes are below 0.00014 V. At 8001 points, changing 40 to 320 longitudinal modes changes these three voltages by less than 10^{-8} V. This bound applies to the tested centerline cases, not to corner fields.

For the experimental case, $I_n = 0.9999999999999$ and $I_p = 0.9999998229$. These independently formed functionals meet the same breakdown condition within the quadrature error. The source jump, reciprocity, dimensions, endpoint voltages and Robin residual are also checked algebraically in the derivation.

8.3 Limiting cases

At $N_d = 0$ the donor term vanishes. As $b \to \infty$, the sidewall becomes insulating and $R_n \to 0$: $\phi \to Vx/L + kx(L - x)/2$, recovering the one-dimensional result. At finite b, all eigenvalues are positive and the mixed-boundary energy identity ensures uniqueness.

9 Physical limitations and reproducibility

The corner at (L, a) remains non-smooth: the bottom Dirichlet trace has zero tangential derivative, while the sidewall Robin condition requires $\phi_y = -V/b$. The transverse-series corner field consequently grows as $(2V/\pi b)\ln M + O(1)$ with mode count M. The Green function does not remove that singularity. This report uses a regular centerline path, and does not take the global maximum field as a device breakdown criterion.

A local oxide boundary is appropriate only for tangential variation lengths large compared with W_I. A Fourier mode of wavenumber p has an oxide correction factor $pW_I \coth(pW_I) = 1 + (pW_I)^2/3 + \cdots$, so rapid end-region variations can violate the approximation. A two-material rounded-end solution is required to predict physical tip fields.

The benchmark assumes fixed depleted charge throughout the rectangle. If neutral material forms, its shape must be determined by a free-boundary or carrier-transport solution. A one-dimensional centerline screening test cannot replace that calculation. Global avalanche must also account for competing paths near the groove, junction and trench tip. The uncomputed high-doping tail is deliberately not presented as a Green function prediction.

The standalone `green_vbd.py` generates the new plots, data and convergence checks. It includes the old A5 basis only for the explicitly labeled comparison panel. `verify_green.py` independently assembles the finite-difference problem and evaluates the transverse Green series. Run both with Python, NumPy, SciPy and Matplotlib. The source archive includes all figure assets required to compile `Green_Function_Breakdown_Report.tex` twice with pdfLaTeX. The OpenDocument file has editable prose and tables; equations are rendered for fidelity and remain editable in LaTeX.

References

[1] I. Pawel, R. Siemieniec and M. Born, *Theoretical Evaluation of Maximum Doping Concentration, Breakdown Voltage and On-state Resistance of Field-Plate Compensated Devices*, ISPS 2008. DOI: 10.1049/ic:20080183. Original Fig. 10 is a simulated characteristic, not reproduced quantitatively here.

[2] W. Zhang et al., *Model and Experiments of Small-Size Vertical Devices With Field Plate*, IEEE Transactions on Electron Devices 66(3), 1416–1421, 2019. DOI: 10.1109/TED.2019.2895328. Fig. 3 supplies the geometry families; Figs. 5 and 7 and Table I supply the schematic interpretation and experiment; Appendix A5 supplies the previous comparison model.

[3] R. van Overstraeten and H. de Man, *Measurement of the ionization rates in diffused silicon p-n junctions*, Solid-State Electronics 13, 583–608, 1970.

[4] Allpix Squared documentation, *Impact Ionization*, original van Overstraeten–de Man parameters, CERN manual. All coefficients used are explicitly listed in this report.